Short Answer Questions

1. Identify the product (s) is / are formed when 1 – methoxy propane is heated with excess HI. Name the mechanism involved in the reaction
2. Draw the major product formed when 1-ethoxyprop-1-ene is heated with one equivalent of HI
3. Suggest a suitable reagent to prepare secondary alcohol with identical group using Grignard reagent.
4. What is the major product obtained when two moles of ethyl magnesium bromide is treated with methyl benzoate followed by acid hydrolysis.
5. Predict the major product, when 2-methyl but -2-ene is converted into an alcohol in each of the following methods.

   (i.) Acid catalysed hydration (ii.) Hydroboration

   (iii.) Hydroxylation using Baeyer's reagent
6. Arrange the following in the increasing order of their boiling point and give a reason for your ordering

   (i.) Butan – 2- ol, Butan -1-ol, 2 –methylpropan -2-ol

   (ii.) Propan -1-ol, propan -1,2,3-triol, propan -1,3 – diol, propan -2-ol
7. Can we use nucelophiles such as NH_3,CH_3O^- for the Nucleophilic substitution of alcohols
8. Is it possible to oxidise t – butyl alcohol using acidified dichromate to form a carbonyl compound.
9. What happens when 1-phenyl ethanol is treated with acidified $KMnO_4$.
10. Write the mechanism of acid catalysed dehydration of ethanol to give ethene.
11. How is phenol prepared from

    i) chloro benzene ii) isopropyl benzene
12. Explain Kolbe's reaction
13. Write the chemical equation for Williamson synthesis of 2-ethoxy – 2- methyl pentane starting from ethanol and 2 – methyl pentan -2-ol
14. Write the structure of the aldehyde, carboxylic acid and ester that yield 4- methylpent -2-en-1-ol.
15. What is metamerism? Give the structure and IUPAC name of metamers of 2-methoxy propane
16. How are the following conversions effected

    i) benzylchloride to benzylalcohol ii) benzyl alcohol to benzoic acid
17. Complete the following reactions

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